

Study Among Undergraduate Students Course Alignment and Personality Fit

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Abstract—This research explores the correlation between personality traits—as defined by the Big Five and Myers-Briggs Type Indicator (MBTI) frameworks—and the academic “efficiency” of students enrolled in BCA and BTech programs. Utilizing a dataset of undergraduate technical students, Exploratory Data Analysis (EDA) was conducted to identify how traits such as Conscientiousness and Introversion influence learning behaviors and adaptability to technical curriculum. Preliminary findings indicate that while curiosity for emerging technologies significantly bolsters confidence, “Anxiety Towards Errors” remains a critical barrier to coding proficiency. This report concludes with actionable recommendations for institutions to implement personality-aligned pedagogical models to enhance student success in technical degrees.

Index Terms—MBTI (Myers-Briggs Type Indicator), EDA (Exploratory Data Analysis), OCEAN (Openness, Conscientiousness, Extroversion, Agreeableness, Neuroticism)

I. INTRODUCTION

With the rapid growth of the IT industry, BCA and BTech degrees have become popular career choices. However, many students struggle due to mismatches between their personality traits and the demands of these programs. This study aims to explore how personality, motivation, and learning behavior impact student performance and satisfaction.

The transition into rigorous technical degrees like BCA and BTech requires more than just mathematical aptitude; it demands a psychological “fit” between the learner and the curriculum. While traditional education focuses on standardized metrics, recent scholarship suggests that personality traits are a primary driver of academic success. For instance, research by Goldberg (1993) highlights that traits like Conscientiousness are often the strongest predictors of long-term academic achievement across various disciplines.

In the specific context of computer science and technical education, the Myers-Briggs Type Indicator (MBTI) has been used to understand how different cognitive styles—such as “Analysts” versus “Explorers”—navigate complex logical problem-solving. Despite the rise of AI and automated tools, the human element—specifically student personality—remains

a constant variable in determining how effectively a student can “Code Without Help” or handle the “Pace of Curriculum.”

This study aims to bridge the gap between psychological theory and technical education by analyzing survey data from students at Lovely Professional University. By examining variables such as “Anxiety Towards Errors” and “Beyond Classroom Learning,” this report seeks to determine if personality can indeed predict the “efficiency” of a student’s journey through a four or three year technical degree.

In the current era of rapidly evolving technology and widespread access to online learning platforms, students pursuing undergraduate technical degrees such as BCA and BTech often face challenges in adapting to the academic structure and skill requirements of these programs. Despite the availability of institutional education, many students struggle with core subjects like programming and mathematics, experience high levels of stress, and lack confidence in their practical abilities.

At the same time, there is an increasing reliance on self-learning through online resources, raising concerns about the effectiveness and relevance of traditional teaching methods. Additionally, a potential mismatch between students’ personality traits, learning styles, and the demands of technical education may contribute to poor academic performance and reduced satisfaction.

Therefore, there is a need to analyze whether current BCA/BTech programs align with student personality and learning behavior, and to identify the key factors affecting student success, motivation, and overall academic experience.

Objectives of the Study

The main objectives of this research are:

- To analyze the relationship between personality traits and academic performance of students in BCA/BTech programs.
- To identify the major challenges faced by students, including difficulties in programming, time management, and academic stress.
- To examine the learning behaviors and preferences of students, particularly the role of self-learning and practical approaches.

- To evaluate whether BCA/BTech courses align with students' personality and career expectations.
- To assess the impact of stress, motivation, and confidence on student performance and engagement.
- To identify the characteristics of students who succeed or struggle in technical education.
- To provide recommendations for improving undergraduate technical courses to make them more effective, practical, and student-centered.

Psychological Framework: Personality Traits and Learning Behavior

Understanding how students learn is not only an academic issue but also a psychological one. Every student has a unique personality, which influences how they think, behave, and respond to learning environments. In this study, two widely recognized psychological models—the Big Five Personality Traits and the Myers-Briggs Type Indicator (MBTI)—are used to understand these differences.

A. The Big Five Personality Traits

The Big Five Personality Traits is a scientific model that explains personality through five broad dimensions. These traits help us understand how a student approaches learning and problem-solving.

1) The Five Traits:

- 1) **Openness to Experience:** Curiosity, creativity, interest in new ideas. Students high in openness enjoy exploring new technologies and concepts.
- 2) **Conscientiousness:** Discipline, organization, responsibility. Students high in this trait follow study schedules and complete assignments on time.
- 3) **Extraversion:** Sociability, communication, energy. Such students prefer group learning and discussions.
- 4) **Agreeableness:** Cooperation, friendliness, teamwork. These students work well in collaborative environments.
- 5) **Neuroticism (Emotional Stability):** Anxiety, stress, emotional sensitivity. High levels may lead to fear of errors and lower confidence.

2) **MBTI (Myers-Briggs Type Indicator):** The MBTI model categorizes individuals into personality types based on how they perceive the world and make decisions. It is widely used for understanding learning preferences and behavior patterns.

MBTI Groups Used in This Study:

- **Analysts** (INTJ, INTP, ENTJ, ENTP): Logical thinkers who prefer problem-solving and independent learning.
- **Diplomats** (INFJ, INFP, ENFJ, ENFP): Creative and intuitive learners who value meaning and interest.
- **Sentinels** (ISTJ, ISFJ, ESTJ, ESFJ): Structured and disciplined learners who prefer organized teaching.
- **Explorers** (ISTP, ISFP, ESTP, ESFP): Practical learners who prefer hands-on and experiential learning.

Impact of Personality on Learning Styles

Both the Big Five traits and MBTI types influence how students interact with learning environments.

Key Connections:

- Students with high openness prefer self-learning, online resources, and exploring new technologies.
- Students with high conscientiousness follow structured study schedules and complete assignments regularly.
- Extraverted students prefer group learning, discussions, and collaborative projects.
- Students with high neuroticism (anxiety) may fear making mistakes and show lower confidence in coding or practical tasks.
- MBTI Analysts are strong in logical problem-solving and independent coding.
- MBTI Explorers prefer hands-on learning and practical exposure.
- MBTI Sentinels perform better in structured classroom environments.

Relevance to This Study

In the context of this research:

- Variables such as curiosity for technology reflect openness.
- Study schedule and assignment completion reflect conscientiousness.
- Group learning preference reflects extraversion.
- Anxiety towards errors reflects neuroticism.
- By linking these traits with academic and behavioral data, the study examines whether personality alignment influences learning effectiveness and employability confidence.

Why This Matters

- Not all students learn in the same way.
- Understanding personality helps explain why some students struggle despite effort, why some prefer practical learning over theory, and why confidence levels differ among students.

II. LITERATURE REVIEW

Previous research has established a significant relationship between personality traits and academic performance. For instance, Goldberg (1993) and Radhika (2024) examined the impact of the Big Five personality traits on students' academic outcomes and found that traits such as conscientiousness and openness positively influence performance. The study highlights that individual differences in behavior, thinking, and emotional patterns contribute to variations in academic success.

However, the study primarily focuses on academic performance measured through grades and does not consider other important factors such as course suitability, learning styles, and student satisfaction. Academic success is not solely dependent on intelligence or personality traits but is also influenced by how well a student's personality aligns with their chosen field of study.

Therefore, the present study extends existing research by analyzing the relationship between personality traits and degree-personality fit, particularly among BCA and BTech students. It aims to explore whether alignment between personality and

coursework influences not only academic performance but also confidence, engagement, and overall career readiness.

III. METHODOLOGY

TABLE I
DATASET DESCRIPTION WITH ENCODING

Attribute	Description	Type	Encoding
Anxiety_towards_errors	Fear of making mistakes	Categorical	1=High, 2=Low, 3=Guidance, 4=Comfortable
Learning_style	Preferred learning method	Categorical	1=Classroom, 2=YouTube, 3=MOOCs
Personality_match_with_program	Alignment with course	Likert	1=Low, 2=Medium, 3=High
Employability_confidence	Job readiness confidence	Likert	1=Low, 2=Medium, 3=High
MBTI	Personality type group	Categorical	1=Analysts, 2=Diplomats, 3=Sentinels, 4=Explorers
Opinion_Category	Thematic response category	Categorical	Coding, Academic, Confidence, Motivation, Practical, Other

A. Data Pre-processing

The collected dataset was preprocessed and cleaned using R programming to ensure accuracy, consistency, and suitability for analysis. The data preparation process involved multiple steps, including handling missing values, transforming categorical responses, and processing open-ended responses.

1) *Handling Missing Values:* Initially, missing values in the dataset were identified using column-wise checks. For numerical variables, missing values were replaced using the mean imputation method, ensuring that no data was lost while maintaining overall distribution consistency.

This approach was applied to variables such as:

- Struggle with subjects
- Personality-course alignment
- Task motivation
- Pace of curriculum
- Personality match with program

This method helped preserve all student responses while minimizing bias due to incomplete data.

2) *Transformation of Categorical Variables:* Since the dataset contained multiple categorical responses (e.g., Yes/No, Agree/Disagree, descriptive options), these were converted into numerical form using Likert-scale encoding.

- Responses such as Yes, No, Unsure and Agree, Disagree were mapped to numeric values (e.g., 1, 2, 3).
- Multi-option responses (e.g., learning styles, study behavior, domain preferences) were also encoded into structured numeric categories.

This transformation enabled:

- Easier statistical analysis
- Compatibility with visualization techniques
- Improved interpretability of relationships between variables

3) *Standardization of Survey Responses:* To maintain consistency:

- Text-based responses were standardized.
- Similar responses were grouped into defined categories.
- Irregular or lengthy responses were simplified into structured formats.

This ensured uniformity across all variables and reduced ambiguity in analysis.

4) *Handling Open-Ended Responses:* The survey included an open-ended question capturing student opinions. These responses were processed using text mining techniques in R:

1) Text was cleaned by:

- Converting to lowercase
- Removing punctuation and numbers
- Eliminating common stopwords

2) A Document-Term Matrix (DTM) was created to extract word frequencies.

3) Frequently occurring words were identified and analyzed.

4) Based on keyword patterns, responses were grouped into meaningful thematic categories, such as:

- Coding Difficulty
- Academic Difficulty
- Lack of Confidence
- Interest/Motivation
- Need for Practical Learning

This process transformed unstructured textual data into structured categories, enabling quantitative analysis.

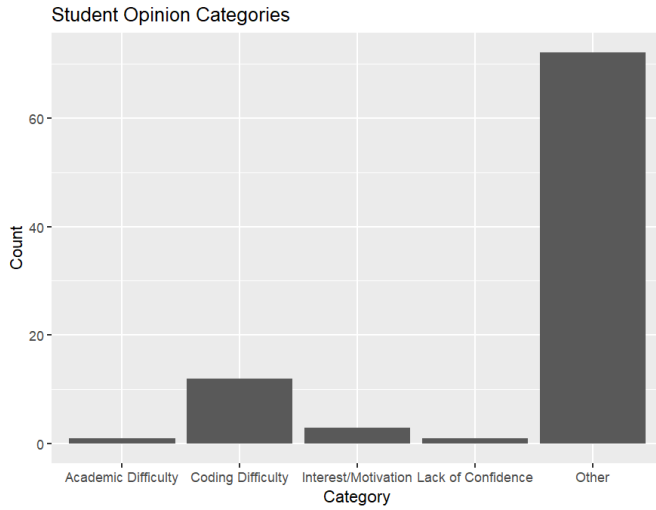


Fig. 1. Opinion bar chart

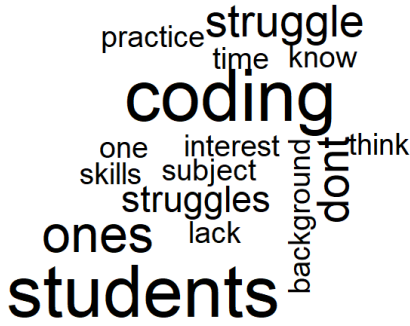


Fig. 2. Frequently appearing words in opinions

B. Data Visualization Preparation

After cleaning and transformation:

- Variables were structured into analysis-ready format.
- Categorical variables were encoded for plotting.
- Data was prepared for visualization using libraries such as ggplot2.

This allowed the generation of meaningful graphs such as:

- 1) Bar charts
- 2) Proportion plots
- 3) Thematic distributions

Tools Used

The following tools and libraries were used for data preparation:

- 1) R Programming Language

- 2) dplyr (data manipulation)
- 3) tm (text mining)
- 4) ggplot2 (visualization)
- 5) wordcloud (text visualization)

Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset, identify patterns, and analyze relationships between variables related to student learning behavior, personality traits, and employability confidence. The analysis was conducted using R programming and visualization techniques (ggplot2).

1) *Data Overview:* The dataset was initially explored using functions such as:

- `str()` to understand structure and variable types
- `summary()` to obtain descriptive statistics
- `sapply()` to verify data types
- `colSums(is.na())` to check missing values

These steps ensured that the dataset was properly cleaned and ready for analysis.

TABLE II
SUMMARY STATISTICS OF DATASET VARIABLES

Variable Name	Type	Min	Q1	Med	Mean	Q3	Max
Student_Age	Num	18	19	20	20.25	21	29
Struggle_with_subjects	Num	1	2	3	2.805	3	5
Practical_confidence	Num	1	3	3	3.27	4	5
Coding_without_help	Num	1	3	3	3.169	4	5
Academic_satisfaction	Num	1	3	3	3.225	4	5
Understanding_memorizing	Num	1	1	1	1.057	1	2
Study_schedule	Num	1	1	3	2.382	3	4
Completing_assignment	Num	1	1	1	1.318	2	2
Anxiety_towards_errors	Num	1	1	3	2.393	3	4
Academicpressure_learning_relation	Num	1	1	2	2.124	3	3
Asking_doubts	Num	1	1	2	1.921	3	3
Group_learning	Num	1	1	2	1.897	3	3
Personality_Course_learning	Num	1	2	3	3.08	4	5
Programming_Student_personality	Num	1	2	3	3.195	4	5
Learning_environment	Num	1	2	2.533	2.533	3	4
Task_motivation	Num	1	3	3	3.193	4	5
Difficult_aspect_of_program	Num	1	3	3.553	3.553	3.553	7
In_face_of_difficulty	Num	1	1	1.913	1.913	2	4
Pace_of_curriculum	Num	1	3	3	3.023	3.023	5
Personality_match_with_program	Num	1	3	3	3.477	4	5
Choice	Num	1	1	1	1.631	2	3
MBTI	Num	1	1	2.078	2.078	2.078	4
Learning_style	Num	1	1	2	1.753	2	3
Student_Domain	Num	1	1	4.463	4.463	8	8
Employability_confidence	Num	1	3	3	3.045	4	5

2) *Demographic Analysis:* Demographic variables such as Student Age and Course Year were analyzed to understand the composition of respondents.

- A histogram was used to visualize the age distribution of students.
- A boxplot was used to examine the relationship between age and course year.
- A jitter plot provided a more detailed view of individual data points across course years.

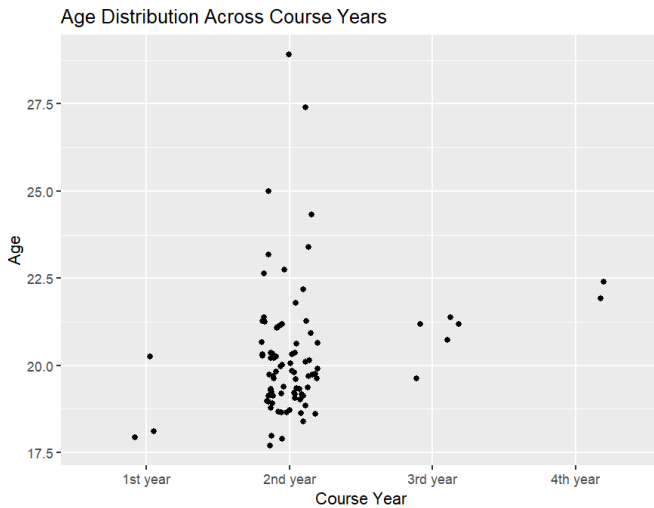


Fig. 3. Age Distribution Across Course Year

Key Insight: The majority of respondents belonged to the 18–23 age group, with a higher concentration of students in the 2nd year, indicating a relatively homogeneous sample.

3) *Visualization Techniques Used:* The following visualization methods were used:

- Histogram → Age distribution
- Boxplot and Jitter Plot → Demographic comparison
- Bar Charts (Grouped and Stacked) → Relationship analysis
- Cross-tabulation tables → Categorical relationships

4) *Summary of EDA Findings:* The EDA revealed several important patterns:

- Curiosity and self-learning significantly may impact employability confidence.
- Study discipline and assignment completion improve technical skills.
- Anxiety and academic pressure negatively affect performance.
- Learning style mismatch leads to increased academic struggle.
- Personality alignment plays a key role in student satisfaction and success.

IV. RESULTS/KEY FINDINGS

A. Relationship Analysis Between Key Variables

Several relationships were analyzed to understand how behavioral, psychological, and academic factors influence student learning and employability confidence.

1) *Curiosity for Technologies vs. Employability Confidence:* A bar chart was used to analyze how curiosity impacts confidence.

Insight: Students with higher curiosity for technologies tend to show higher employability confidence, indicating the importance of self-driven learning. (EDA shows correlation, not causation.)

10. I enjoy learning new technologies on my own.
89 responses

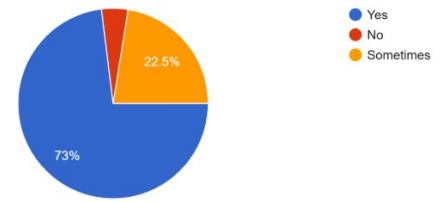


Fig. 4. Curiosity towards technology

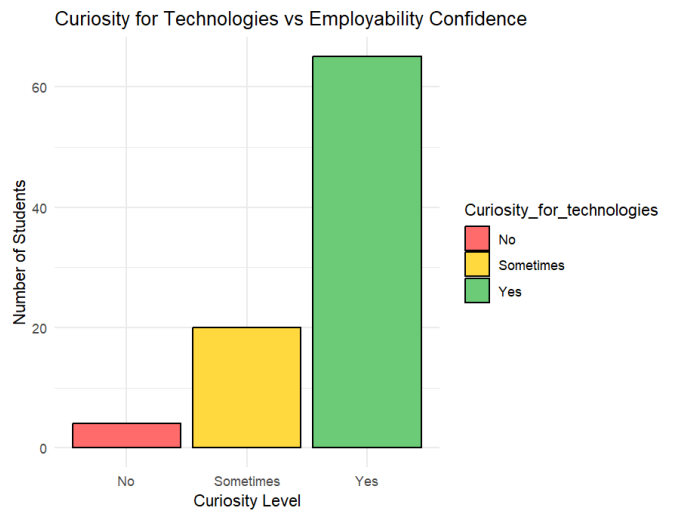


Fig. 5. Correlation between curiosity towards technology vs. employability confidence.

2) *Study Schedule vs. Employability Confidence:* A grouped bar chart was used to compare study habits and confidence levels.

Insight: Students who maintain a consistent study schedule demonstrate higher employability confidence, emphasizing the role of discipline in learning outcomes.



Fig. 6. Correlation between study schedule pattern vs. employability confidence

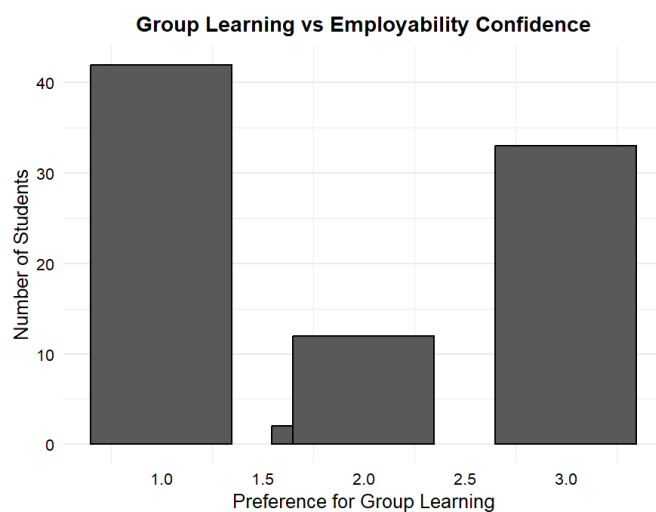


Fig. 8. Group learning vs. employability confidence.

12. I follow a regular study or practice schedule.
89 responses

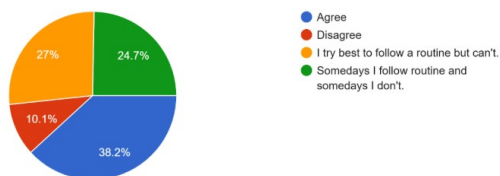


Fig. 7. Pie chart for study pattern

17. I enjoy group projects and collaborative learning.
87 responses

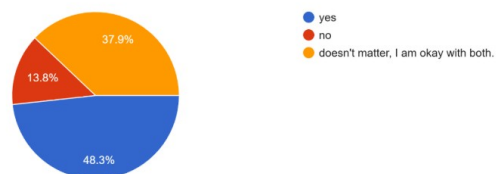


Fig. 9. Group learning survey responses

3) *Group Learning vs. Employability Confidence:* The relationship between collaborative learning and confidence was analyzed.

Insight: Students who engage in group learning tend to exhibit higher confidence, suggesting that collaborative environments enhance understanding and skill development.

4) *Anxiety Towards Errors vs. Employability Confidence:* This analysis examined the psychological impact of fear of mistakes.

Insight: Higher anxiety levels are associated with lower employability confidence, indicating that fear of failure negatively affects learning and performance.

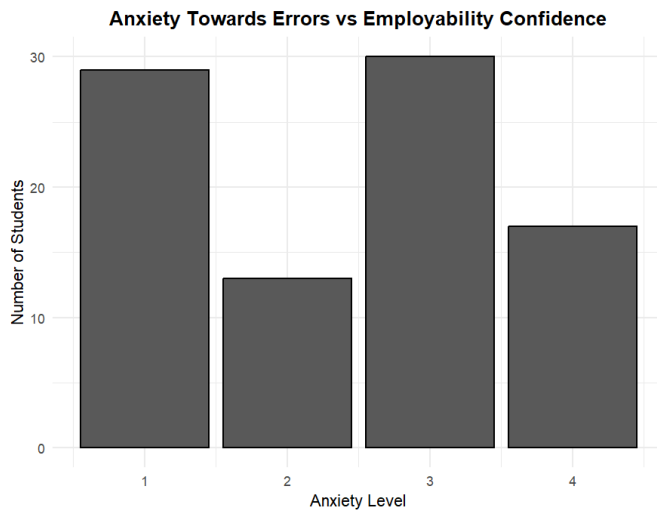


Fig. 10. Correlation between anxiety towards errors and employability confidence.

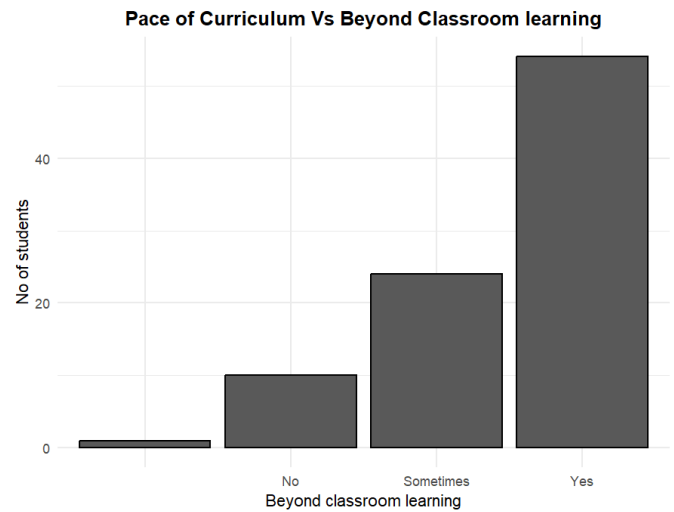


Fig. 13. Correlation between pace of curriculum and beyond classroom learning.

14. I feel anxious when debugging or facing errors.
89 responses

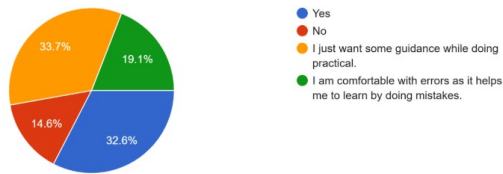


Fig. 11. Anxiety among student survey response.

5) *Beyond Classroom Learning vs. Pace of Curriculum:*
This analysis explored whether curriculum pace influences external learning.

Insight: Students experiencing a fast-paced curriculum are more likely to engage in learning beyond the classroom, suggesting gaps in formal education delivery.

25. I feel overwhelmed by the pace of the BCA/Btech curriculum
88 responses

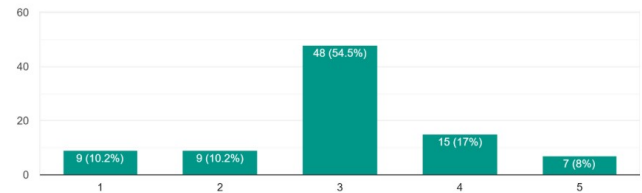


Fig. 14. Survey response towards pace of curriculum

11. I experiment with tools or code beyond what is taught in class.
88 responses

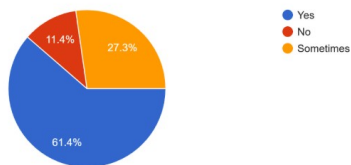


Fig. 12. Beyond classroom learning survey

6) *Struggle with Subjects vs. Learning Style:* The relationship between learning style and academic difficulty was analyzed.

Insight: Students whose learning style does not align with teaching methods tend to struggle more, highlighting the need for adaptive teaching strategies.

30. Which learning style do you prefer?
89 responses

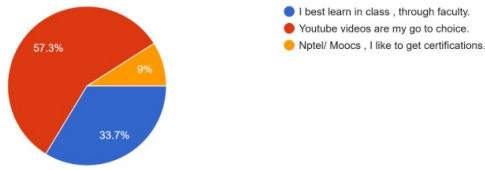


Fig. 15. Learning style survey responses.

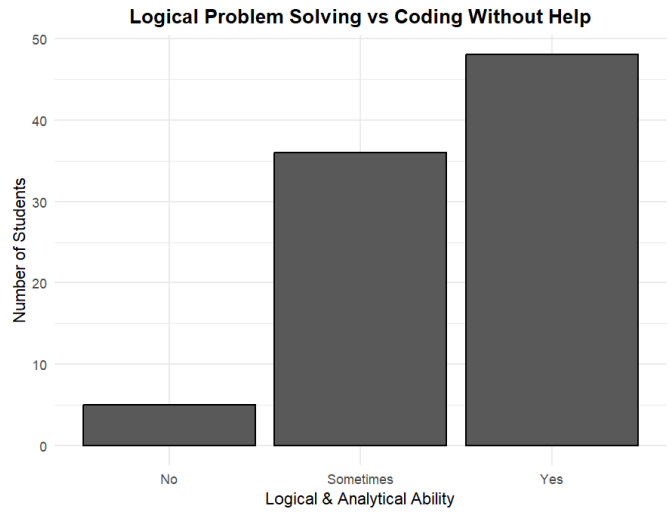


Fig. 17. Correlation between logical problem solving and coding without help.

8) *Personality Match with Program vs. Academic Satisfaction*: This analysis evaluated the alignment between student personality and chosen program.

Insight: Students whose personality aligns with their program report higher academic satisfaction, indicating that personality fit plays a crucial role in educational success.



Fig. 16. Correlation between struggle with subject and learning style

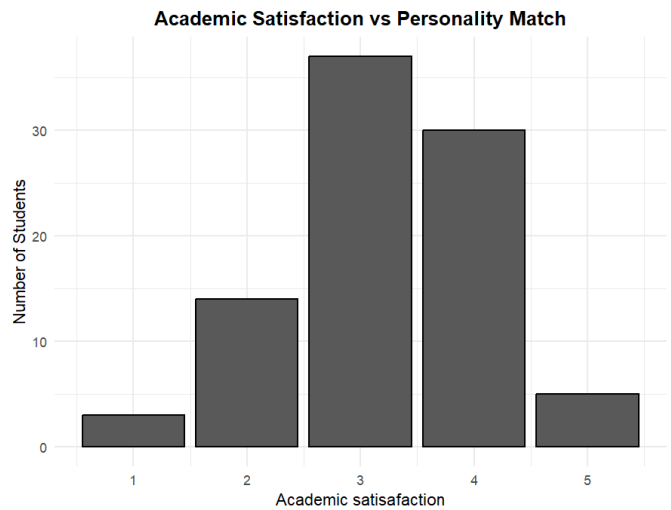


Fig. 18. Correlation between academic satisfaction and personality match.

7) *Logical Problem Solving vs. Coding Without Help*: This relationship examined the importance of analytical ability in programming.

Insight: Students with strong logical and analytical skills are more capable of coding independently, emphasizing the importance of problem-solving in technical education.

V. DISCUSSION

The findings of this study highlight that student learning outcomes in undergraduate technical education are influenced by a combination of academic, behavioral, and psychological factors.

The analysis showed that students with higher curiosity for technologies tend to exhibit greater employability confidence. This aligns with the concept of openness from the Big Five

personality traits, where individuals who are more curious and willing to explore new ideas demonstrate better adaptability and learning outcomes.

Similarly, students who maintain a consistent study schedule showed higher levels of confidence, indicating the role of conscientiousness, which is associated with discipline and goal-oriented behavior. This suggests that structured learning habits significantly contribute to academic success.

The study also found that group learning positively impacts student confidence and understanding. This can be linked to extraversion, where individuals benefit from interaction and collaborative environments.

On the other hand, anxiety towards errors was observed to negatively impact employability confidence. This reflects the role of neuroticism, where higher levels of stress and fear can hinder learning and performance. Students who are afraid of making mistakes may avoid practical engagement, especially in programming tasks.

The results further indicate that a mismatch between learning styles and teaching methods leads to increased struggle among students. This suggests that the current education system may not adequately address diverse learning preferences, particularly for students who prefer hands-on or self-directed learning approaches.

Additionally, the relationship between logical problem-solving ability and coding skills emphasizes the importance of analytical thinking in technical education. Students with stronger logical skills demonstrated better ability to code independently.

The findings also highlight that personality alignment with the chosen academic program significantly influences academic satisfaction. Students whose interests and personality traits match their field of study tend to perform better and experience less stress.

Overall, the study suggests that a one-size-fits-all approach in education is ineffective, and there is a need for more flexible, personalized, and practical learning environments.

A. Limitations

While the findings provide valuable insights into the intersection of personality and technical education, the following limitations must be acknowledged:

- **Sample Size and Diversity:** The current study relies on a sample size of approximately 80 respondents. While sufficient for exploratory data analysis (EDA), a larger and more geographically diverse sample would be required to generalize these findings across the entire BCA/BTech student population in India.
- **Self-Reported Bias:** Data regarding CGPA, personality traits (MBTI/Big Five), and learning preferences were self-reported. This introduces the risk of Social Desirability Bias, where students may over-report positive traits or academic performance.
- **Subjectivity of “Efficiency”:** The term “degree efficiency” is multifaceted. This study primarily focuses on the alignment of personality with curriculum pace and

employability confidence, which may not capture other efficiency metrics like technical skill proficiency or long-term career growth.

- **Psychometric Limitations:** The use of simplified MBTI and Big Five assessments provides a thematic overview but lacks the depth of clinical psychological testing.

VI. CONCLUSION

Academic and Learning Behavior

- Students who follow a consistent study schedule tend to have higher employability confidence.
- Assignment completion is positively related to independent coding ability.
- Students with strong logical and analytical skills perform better in coding without help.

Personality and Psychological Factors

- Students with high curiosity for technologies show greater employability confidence.
- Personality match with the program significantly influences academic satisfaction.
- Students experiencing high anxiety towards errors tend to have lower confidence levels.

Learning Environment and Style

- Group learning positively contributes to confidence and understanding.
- Mismatch between learning style and teaching method leads to increased struggle in subjects.
- Students with hands-on or self-learning preferences often rely on external learning sources.

Curriculum and System Limitations

- A fast-paced curriculum forces students to learn beyond the classroom.
- Many students reported difficulty in programming, lack of guidance, and need for practical learning.
- Open-ended responses highlighted issues such as: coding difficulty, lack of confidence, and need for more practical exposure.

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